

$$28. \begin{cases} \dot{x}_1 = x_2 \end{cases}$$

$$\begin{cases} \dot{x}_2 = \vartheta_1 \sin x_1 + \vartheta_2 \cos x_1 + u \end{cases}$$

• условие: $\lim_{t \rightarrow \infty} e(t) = 0$

$$e(t) = x_m(t) - x(t)$$

• $\vartheta_1 = 1, \vartheta_2 = 2$; $\hat{\vartheta}_1(0) = \hat{\vartheta}_2(0) = 0.5$

ϑ -неизвестные

• нормальная модель:

$$\begin{cases} \dot{x}_{m1} = x_{m2} \end{cases}$$

$$\begin{cases} \dot{x}_{m2} = \underbrace{-\delta}_{\tilde{a}_{m0}} x_{m1} - \underbrace{6}_{\tilde{a}_{m1}} x_{m2} + \delta g \end{cases}$$

$$g = 2\delta \sin b + 2b'$$

• $\dot{e} = A_m x_m + b_m g - \dot{x}$

$$\begin{cases} \dot{x}_1 = x_2 \end{cases}$$

$$\begin{cases} \dot{x}_2 = a_{m0} x_1 + a_{m1} x_2 + \vartheta_1 \sin x_1 + \vartheta_2 \cos x_1 + u \end{cases}$$

$$\dot{x} = A_m x + E_2 (\tilde{a}_{m0} x_1 - \tilde{a}_{m1} x_2 - a_{m0} x_1 - a_{m1} x_2 + \vartheta_1 \sin x_1 + \vartheta_2 \cos x_1 + u)$$

$$E_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$A_m = \begin{pmatrix} a_{m0} \\ a_{m1} \end{pmatrix}$$

$$\dot{x}_m = A_m x_m + b_m g$$

$$A_m = \begin{pmatrix} 0 & 1 \\ -\delta & -6 \end{pmatrix}, b_m = \begin{pmatrix} 0 \\ \delta \end{pmatrix}$$

Отсюда

$$\dot{e} = Am \dot{x}_m + b \omega g - Am x - E_2 (Am^T x + \vartheta_1 \sin x_1 + \vartheta_2 \cos x_1 + u)$$

$$\dot{e} = Am e + E_2 (fg - Am^T x - \vartheta_1 \sin x_1 - \vartheta_2 \cos x_1 - u)$$

$$u = fg - Am^T x - \vartheta_1 \sin x_1 - \vartheta_2 \cos x_1$$

$$\vartheta \rightarrow \hat{\vartheta}$$

$$u = fg - Am^T x - \hat{\vartheta}_1 \sin x_1 - \hat{\vartheta}_2 \cos x_1$$

подставим в уравнение системы:

$$\begin{aligned} \dot{e} &= Am e + E_2 (fg - Am^T x - \hat{\vartheta}_1 \sin x_1 - \hat{\vartheta}_2 \cos x_1) \\ &= fg + Am^T x + \hat{\vartheta}_1 \sin x_1 + \hat{\vartheta}_2 \cos x_1 \end{aligned}$$

$$\dot{e} = Am e - E_2 \tilde{\vartheta}^T R = Am e + E_2 \tilde{\vartheta}^T (-R)$$

$$R = \begin{pmatrix} \sin x_1 \\ \cos x_1 \end{pmatrix} \quad \tilde{\vartheta} = \vartheta - \hat{\vartheta}$$

$$\dot{e} = Ae + b \tilde{\vartheta}^T \omega \Rightarrow \dot{\tilde{\vartheta}} = \gamma \omega b^T P e \Rightarrow \boxed{\dot{\tilde{\vartheta}} = -\gamma^* R E_2^T P e}$$

$$A_m^T P + P A_m = -Q, \quad Q \succ 0$$